nonlinear and linear systems are in fact *topologically conjugate*: if z is the solution of the nonlinear system and δz is the solution of the linear system, then there is a smooth, invertible map $\boldsymbol{\Phi}$ of $\mathbb{R}^m$ defined in a neighborhood of the origin such that

$$z(t) = z^* + \boldsymbol{\Phi}(\delta z(t)).$$

This is referred to as the Hartman-Grobman theorem (for more discussion, see [177], where this result is referred to as the "Linearization Theorem"; a proof may be found in [362]).

Let z^* be an equilibrium point. We say that z^* is stable ("in the sense of Lyapunov") if, for all ε, there exists δ such that, for all z_0 such that $\|z_0 - z^*\| < \delta$,

$$\sup_{t \geq 0} \|\mathscr{F}_t(z_0) - z^*\| < \varepsilon.$$

That is, the solution started near to z^* stays near z^* for all time. The Hartman-Grobman theorem clearly implies that the stability of a given hyperbolic equilibrium point z^* of a nonlinear system can be inferred from the stability of the origin for the linearization of the system around z^*.

In the case of Hamiltonian systems, the eigenvalues are, very often, on the imaginary axis, and the Hartman-Grobman theorem is not of much use. However, there is another result which is relevant in this case. Observe that an equilibrium point $z^* = (\boldsymbol{q}^*, \boldsymbol{p}^*)$ of a Hamiltonian system in "kinetic plus potential" form

$$H(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q})$$

will always have $\boldsymbol{p}^* = 0$ and $\nabla U(\boldsymbol{q}^*) = 0$. We say that $\boldsymbol{q}^*$ is a *strong local minimum* of the potential if there exists $\varepsilon > 0$ such that

$$0 < \|\boldsymbol{q} - \boldsymbol{q}^*\| < \varepsilon \Rightarrow U(\boldsymbol{q}) > U(\boldsymbol{q}^*).$$

Then we have

Theorem 1.1 *If $\boldsymbol{q}^*$ is a strong local minimum of smooth potential U then $z^* = (\boldsymbol{q}^*, \boldsymbol{0})$ is stable.*

The proof of this theorem relies on showing that if trajectories are started from a point sufficiently close to z^* they cannot wander away to infinity. Although the result holds in greater generality, it is easy to show under assumptions of local smoothness of U (which we are normally happy to make in molecular dynamics). For more discussion see the text [216]. If the potential is C^2, the linearized version of this system has "kinetic plus potential" form with Hamiltonian

$$\bar{H} = \frac{\delta \boldsymbol{p}^T \boldsymbol{M}^{-1} \delta \boldsymbol{p}}{2} + \frac{\delta \boldsymbol{q}^T U''(\boldsymbol{q}^*) \delta \boldsymbol{q}}{2}.$$